

Jayadratha Gayen

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Python and AI Engineer with production experience in LLM-powered agentic systems, graph neural networks, and 3D computer vision. Currently building agentic coding assistants using LangGraph, MCP, and RAG; prior research spans temporal GNNs, abstention-based graph classification, and 3D motion transfer using GCNs. Published researcher with work in legal AI, medical graph learning, and biological neural modeling.

Technical Skills

Languages: Python, SQL, C++

GenAI / Agentic AI: LangGraph, LangChain, MCP, Agentic AI, LLM APIs, RAG, Tool Use, LLM Evaluation

ML / DL: PyTorch, PyG, TensorFlow, Transformers, HuggingFace, scikit-learn, Graph Neural Networks, Computer Vision

Infrastructure: AWS, Docker, Git, MongoDB, Neo4j, Vector Databases, Knowledge Graphs

Work Experience

CloudAngles Inc.

Feb 2026 – Present

AI Research Engineer | Hyderabad, India

- Built core GenAI components for Codebenders, an agentic coding assistant: designed repository indexing with semantic retrieval, context-management pipelines, and tool orchestration using LangGraph and LLM APIs for natural language code editing, debugging, and generation.
- Reduced AI token usage by **76.3%** and API costs by **64.2%** through in-session tool result caching and smarter tool loading, keeping the agent cost-efficient at scale.
- Improved agent response speed by **22.4%** by reworking context window management and session memory, so the agent picks up exactly where it left off after any interruption.
- Built automated **LLM evaluation pipelines** with custom benchmarks for code quality, task completion rate, and model reliability across real developer scenarios.
- Worked with MCP to standardize tool interfaces and reduce integration overhead across the agentic system.

Google Summer of Code at INCF

May 2025 – Sep 2025

Open Source Developer | Remote

- Built **DevoTG** in Python: a temporal graph neural network framework for modeling *C. elegans* neural development as evolving graphs; temporal approach achieved mean test AUC of **0.839 vs 0.577** for static methods.
- Designed dataset pipelines for spatiotemporal biological data and built interactive **3D visualizations**.

Machine Learning Lab (MLL), IIIT Hyderabad

Nov 2022 – Jul 2025

Research Associate | Hyderabad, India

- Designed and trained graph neural network models in Python and PyTorch for node classification with uncertainty management, applied to legal judgment prediction and disease diagnosis; published at **TMLR 2025**.
- Built abstention frameworks for continuous time dynamic graphs (CTDGs), improving AUC/AP by **10–15%** under class imbalance; published at **ASONAM 2025** and **KDD TGL Workshop 2025**.
- Developed end-to-end ML pipelines in Python using PyTorch and PyG across legal, medical, and biological graph datasets, covering training, evaluation, and visualization.
- Worked on 3D computer vision research for motion transfer across diverse character topologies (humanoids and quadrupeds) using graph convolution networks; outperformed standard pose transfer methods on the Mixamo dataset.

Vikram Solar Limited

Jul 2018 – Dec 2019

Engineer | Falta, West Bengal, India

- Led a Six Sigma throughput improvement project, increasing production by **10%** through bottleneck analysis using DOE, Pareto, and Fishbone methods.
- Introduced RFID automation and SAP-based data analytics workflows to reduce manual errors in manufacturing.

Education

International Institute of Information Technology Hyderabad

M.Sc in Electronics and Communication Engineering by Research (CGPA: 8.29/10)

2022 – 2025

Indian Institute of Engineering Science and Technology Shibpur

Bachelor of Technology in Electrical Engineering (CGPA: 8.02/10)

2014 – 2018

Selected Publications

Node Classification With Reject Option (TMLR 2025)

- Proposed **NodeCwR**, a reject-option GNN framework that abstains on uncertain predictions to improve reliability in high-stakes domains like legal and medical applications.

Reducing Misclassification Risk in Dynamic Graph Neural Networks Through Abstention (ASONAM 2025)

- Introduced abstention mechanisms for temporal GNNs, improving AUC/AP by **10–15%** while handling class imbalance and critical misclassifications.

DevoTG: Temporal GNNs for Modeling C. elegans Developmental Connectomics (arXiv 2026)

- Built a temporal GNN framework for modeling neural development; mean test AUC of 0.839 vs 0.577 for static methods.

Selected Projects

Nyay Saheli: Conversational Legal Assistant (LLM + GraphRAG)

2025

- Built a multilingual legal AI assistant using **RAG** and **GraphRAG** for legal guidance and question answering in Indian languages; National Finalist (Top 6/275 teams) at **BharatGen Hackathon**.
- Developed OCR ingestion, embedding, hybrid search, reranking, and retrieval pipelines over legal knowledge bases using vector databases and knowledge graphs.

Node Classification with Reject Option for Legal Judgement Prediction (NLP + GNN)

2024

- Built NodeCwR, a GNN framework with cost-based and coverage-based rejection for Legal Judgment Prediction, analyzing complex legal texts using NLP and graph-based representations; achieved 6–19% accuracy improvement on the ILDC dataset.

3D Motion Transfer Across Diverse Character Topologies (GCN + 3D Vision)

2023

- Built a GCN-based method for realistic 3D motion transfer between diverse character topologies (humanoids and quadrupeds); used prior frame encodings to reduce jitter and outperformed standard pose transfer methods on the Mixamo dataset.

Multilingual News Similarity Detection (Transformers + NLP)

2022

- Built a multilingual document similarity system using Siamese Transformer architectures; best result with **multilingual DistilBERT** (PCC: 0.5683) in SemEval 2022 Task 8.

Leadership & Activities

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- National Finalist (Top 6/275 teams), **BharatGen Hackathon** – GenAI multilingual legal assistant.
 - Teaching Assistant, **3D Vision Summer School**, IIIT Hyderabad – conducted tutorials on graph neural networks and held doubt sessions for participants from across India.